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TAXATION IN EMPIRICAL LABOUR SUPPLY MODELS: LONE MOTHERS IN THE UK*

Richard Blundell, Alan Duncan and Costas Meghir

Taxation has become an integral part of the empirical specification of labour supply models. The opportunity set available to workers is made significantly non-linear by the workings of the tax (and benefit) system and this, in turn, complicates an individual's optimal choice of labour supply. In such a framework the relationship between hours of work, gross wages and other income is necessarily non-linear. Moreover, the marginal wage and virtual income at any chosen hours of work are an implicit function of the hours choice itself. To guarantee a unique solution to this choice problem consumer theory places integrability conditions on the form of wage and income derivatives of labour supply. Similarly, once the problem is made stochastic and econometric methods are utilised to recover unknown parameters, a coherency condition is required to define a well behaved econometric model (see Heckman (1978) and Gourieroux *et al.* (1980)). In the labour supply and taxation framework these two conditions are quite naturally closely related. They do, however, have implications for labour supply specification and the reliability of tax reform simulations based on empirical labour supply models.

For the case of the linear labour supply model, MaCurdy *et al.* (1990) document the potential for misleading results that may occur once a linear model is utilised in a framework with non-linear taxation. Such a specification all but guarantees a global forward sloping labour supply function which is not a requirement of economic theory and is empirically often too restrictive. In this paper we focus on two issues. The first relates to the specification of a flexible model allowing backward bending labour supply. The second issue relates to the statistical coherency of such a model in the presence of non-linear taxation.

The empirical application in this paper concerns the labour supply behaviour of lone mothers in the UK. This group face a highly non-linear budget constraint. The analysis of their labour supply responses is crucial for the design of effective policy reforms. We use a labour supply specification suggested by Heckman (1974) applied to a sample of data drawn from six years of the *Family Expenditure Survey*.

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I. LABOUR SUPPLY SPECIFICATION

For tax policy analysis labour supply specifications should be sufficiently flexible, whilst at the same time maintaining theory consistency over a wide range of hours/wage combinations (see Blundell *et al.* (1988) and Stern (1986)). With these requirements in mind, we have chosen a specification introduced by Heckman (1974). Let the marginal rate of substitution (MRS) between goods and non-work time be

$$\ln(m) = \alpha + \beta h + \gamma Y^*, \tag{1}$$

where m is the marginal rate of substitution, h is hours of work and Y^* is a utility level indicator. To define the latter consider an individual with hourly wage rate w and other income μ. Then Y^* represents that level of consumption at which the individual is indifferent between not working (and receiving Y^*) and working to a point h^* to generate income $c^* = wh^* + \mu$. By choosing a log-linear specification, the MRS remains positive for all possible h and Y^*. For a diminishing MRS between consumption and non-work time (convexity of indifference curves), $\partial m/\partial h > 0$ which is guaranteed for $\beta > 0$. Equating (1) to the log real wage, the MRS function defines the implicit labour supply function by

$$\ln(w) = \alpha + \beta h + \gamma\left[wh + \mu - \frac{w}{\beta}(1 - e^{-\beta h})\right] + u, \tag{2}$$

where the ordinal utility level indicator is given by $Y^* = wh + \mu - w(1 - e^{-\beta h})/\beta$ and where u represents random preference variation entering through α.[1] As shown in Heckman (1974), the wage and virtual income derivatives $\partial h/\partial w$ and $\partial h/\partial \mu$ of (2) are

$$\frac{\partial h}{\partial w} = \frac{(1/w) - \gamma[h - (1 - e^{-\beta h})/\beta]}{\beta + \gamma w(1 - e^{-\beta h})} \tag{3}$$

and

$$\frac{\partial h}{\partial \mu} = \frac{-\gamma}{\beta + \gamma w(1 - e^{-\beta h})}. \tag{4}$$

The MRS specification is consistent with economic theory when $\beta > 0$ and $\gamma > 0$, restrictions that are not data dependent. As can be seen from (3) and (4) this does not restrict the labour supply function to be forward sloping at high hours.

The implicit labour supply function (2) can be made consistent with intertemporal optimisation under uncertainty[2] by defining $\mu = c - wh$, c being observed household consumption and w the after-tax marginal wage rate (see, for example, Blundell and Walker (1986)). Finally note that observed heterogeneity in preferences, induced by the presence of children and age, can be controlled for by making the parameters of the MRS depend on the demographic characteristics z. In this framework other income μ is a choice variable and will probably be correlated with u. Moreover both the pre-tax

[1] Since $\ln(m)$ is unbounded, no bounds are required on the distribution of u.

[2] With preferences being additively separable across time.

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wage rate and the marginal tax rate may be endogenous. We discuss identification at the end of section II.

II. PIECEWISE LINEAR BUDGET SETS AND STATISTICAL COHERENCY

Consider, for the purposes of illustration, a linear labour supply model

$$h = \alpha + \beta w + \gamma \mu + u \tag{5}$$

where u represents random preference variation. With non-linear taxation the marginal wage and other income variables become non-linear functions of working hours. The implementation of a maximum likelihood estimator requires a coherency condition that ensures the existence of a reduced form solution for the model.

Imagine a single kink point in the budget constraint at h^t defined by a progressive tax structure and suppose that the budget constraint is convex and piecewise linear. If we let w^t be the marginal wage above h^t and μ^t the corresponding virtual income, then (following Hausman (1985)) we may define the probability of desired hours being on the kink h^t by

$$\begin{aligned}\Pr(\alpha+\beta w^t+\gamma\mu^t+u \leqslant h^t \leqslant \alpha+\beta w+\gamma\mu+u) \\ = \Pr(h^t \leqslant \alpha+\beta w+\gamma\mu+u) - \Pr(h^t \leqslant \alpha+\beta w^t+\gamma\mu^t+u),\end{aligned} \tag{6}$$

where $w^t = (1-t)w$ and $\mu^t = \mu + twh^t$. For coherency we require that (6) is non-negative and that the sum of probabilities is unity which is equivalent to $\beta - \gamma h^t > 0$, precisely the Slutsky condition evaluated at h^t. Notice that if leisure is a normal good γ is negative and as a result a kink point (or corner solution) near zero hours ensures $\beta > 0$, or forward sloping labour supply behaviour everywhere.

In this extreme case of the linear labour supply model one kink is therefore sufficient to restrict the implied behaviour over the entire space of wages and hours. As can be seen from the above discussion such a result can be obtained just by considering the imposition of the reservation wage condition at the non-participation kink: for coherency at that point we require forward sloping labour supply which, with a linear specification, implies a positive wage derivative everywhere.

The linear labour supply example illustrated the close link between statistical coherency and the economic theory integrability conditions. In fact this link extends to any model of labour supply however non-linear. Consider the general condition for coherency $h(w^t, \mu^t) \leqslant h(w, \mu)$. A first order Taylor series expansion around $t = 0$ produces the condition $\partial h/\partial w - h^t(\partial h/\partial \mu) \geqslant 0$ which is exactly the standard Slutsky condition. The sufficiency result is also easy to show. In other words the economic integrability conditions at the kink points become an identifying assumption.

For coherency not to affect the global properties of the labour supply function we need a sufficiently flexible specification. Such a specification will have to contain more wage and other income parameters than kinks in the budget constraint since each kink will impose a separate set of restrictions on

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these parameters. A flexible estimation strategy allowing for these issues is described below.

Before we turn to estimation note that the discussion of the probability (6) relates to desired labour supply. If all random variation can be attributed to individual preference heterogeneity there would be mass points in the data space at the kinks h^t. A log likelihood function which would account explicitly for these kinks would automatically impose coherency conditions. This is because the likelihood would be a function of the log of this probability for each individual observed on the kink; a probability that tended to zero would thus act as a penalty function against parameters in the region of the parameter space where coherency is not satisfied. Although clustering associated with kinks in the budget constraint is evident in the United Kingdom (see Duncan (1991), for example), measurement and optimisation errors are bound to be present. These smooth out precise mass points (see Burtless and Hausman (1978)) by dispersing the data around the kinks. Hence more realistically one would have to account for such optimisation errors. In this case the log likelihood function involves weighted sums of such probabilities for all individuals (see Hausman (1985)). Some of these probabilities may be non-positive without the weighted sum being negative or zero, resulting in a likelihood function which does not contain an automatic penalty function. The estimated parameters in this case could lie outside the coherency region.

II.1. *Estimation Methodology*

The British tax system for the years covered by our data is characterised by many kinks, non-convexities and discontinuities as a result of the benefit system and the National Insurance system. Nevertheless an important feature of the tax system is that it is essentially linear to the right of the point at which individuals start paying the standard rate of income tax. Although there are further kinks at higher earnings levels we observe practically no women there. Hence most of the problems relating to the effects of imposing coherency on labour supply models are due to observations below the standard rate of tax.[3] Our estimation strategy is thus very simple: we condition on women who pay the standard rate of tax and estimate our model over the linear part of the budget constraint. Hence the shape of the budget constraint is not allowed to have a direct influence on the labour supply estimates through the imposition of coherency conditions at kink points.

More formally we define the conditional expectation

$$E(u \mid I > 0, \mathbf{z}, \mathbf{x}, w, \mu) = g(w, \mu, \mathbf{z} \mid \theta) + \rho_\lambda \lambda(\mathbf{x}) + \rho_w \epsilon_w + \rho_\mu \epsilon_\mu, \tag{7}$$

where $g(.)$ is the implicit labour supply function defined in the previous section, I is the selection index, $\mathbf{z}$ are the included exogenous variables and $\mathbf{x}$ is the complete set of conditioning variables in the model. The three additional terms on the right hand side are residuals controlling for the selection mechanism, the endogeneity of wages and the endogeneity of other income. In general $\lambda(\mathbf{x})$ is

[3] We also remove individuals paying tax and simultaneously receiving benefits but there are relatively few such cases.

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a function of the probability of selection which can be estimated non-parametrically. In the special case where the index function I determining selection and the labour supply function are jointly normally distributed $\lambda(\mathbf{x})$ is the inverse of Mill's ratio. Although the non-parametric approach would be fully consistent with our structural model it does require vary large datasets and hence we impose joint normality here and we test it. This requires us to assume a particular reduced form equation for the selection equation which we denote by $I^* = \boldsymbol{\beta}_I' \mathbf{x}_I + \epsilon_I$.[4] Finally we also define reduced form equations $\ln w = \boldsymbol{\beta}_w' \mathbf{x}_w + \epsilon_w$ for the log wage and $\mu = \boldsymbol{\beta}_\mu' \mathbf{x}_\mu + \epsilon_\mu$ for other income. We assume for simplicity that all errors $(u, \epsilon_I, \epsilon_w, \epsilon_\mu)$ are jointly normal.[5]

Given identification, which we discuss below, our estimation strategy is as follows. We first estimate the reduced form wage equation corrected for selectivity and the other income equation and construct the residuals $\hat{\epsilon}_w$, $\hat{\epsilon}_\mu$. Conditional on these we define the likelihood function for hours of work and for the selection mechanism as

$$L_1 = \prod_{I^* \leqslant 0} \Pr(I^* \leqslant 0) \cdot \prod_{I^* > 0} \left\{ \Pr(I^* > 0 \mid u) f(u) \left| \frac{\partial u}{\partial h} \right| \right\} \tag{8}$$

with $f(u)$ representing the conditional density of u given ϵ_w *and* ϵ_μ and $\partial u / \partial h$ denoting the Jacobian of transformation from u to h.[6] This empirical strategy controls for endogeneity of the pre-tax wage rate, for the endogeneity of other income and most importantly for non-linear taxes. In effect our empirical strategy identifies labour supply behaviour by estimating our model over a region of the data where the coherency conditions are automatically satisfied (due to the absence of kinks over that region). Our approach allows us to test integrability restrictions without inducing failure of the coherency condition.[7] Note that as shown in Smith and Blundell (1986) we can test for the exogeneity of the pre-tax wage rate and of other income by testing whether the coefficients attached to the respective residuals are zero. The conditional likelihood contains pre-estimated parameters and we correct the standard errors for this.

The estimation strategy we have described is valid when the only source of stochastic variation is random preferences. If we allow for optimisation errors, there will be some individuals to the right of the kink who would have chosen

[4] In estimating both on the whole sample of workers and on the selected sample of those working on the flat part of the budget constraint we use different but always linear selection rules and impose joint normality. As a result the two models are not formally nested and we do not treat them as such.

[5] The bivariate normality assumption can be relaxed by allowing polynomial terms and interactions in the reduced form equations and by using polynomials of the residuals ϵ_w and ϵ_μ in the labour supply equation. Similarly for the selectivity term we would need to include conditional expectations (given $I > 0$) of the selection index residual (see Lee (1982)).

[6] The Jacobian here is $\partial u / \partial h = -\{\beta - \gamma w[1 - \exp(-\beta h)]\}$. For the likelihood function to be well-defined, the Jacobian must be non-vanishing. This bounds β and γ away from zero. Otherwise this does not impose any serious restrictions on the parameters since the Jacobian will only be zero at particular data points that occur with probability zero.

[7] An alternative empirical strategy could have been to use a sufficiently rich preference structure so that the coherency restrictions at the kinks would not impose restrictions on the shape of labour supply elsewhere. In a random preference framework this would require modelling the mass points at the kinks. Extrapolating our model over the whole sample requires the assumption that the selected data reflect preferences of all individuals.

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to work on the segment to the left. For these the marginal tax rate will be measured with error. There are two potential solutions to this. First, we instrument the marginal wage rate even after selecting on the flat part of the budget constraint. More importantly we can mitigate the problem by selecting deeper into the tax segment. The larger our sample the further away from the kink we can select. If the optimisation error has finite variance and the segment to the right of the kink is sufficiently large this procedure will asymptotically purge all the measurement error. In our empirical work we tried this selection procedure but we did not obtain significantly different results.

The identification of our model requires at least three exclusion restrictions from the implicit labour supply equation: one for the selectivity term, one for the wage rate and one for other income. We exclude from the labour supply equation the demand side variables (regional unemployment, regional vacancies, regional redundancies and female unemployment by age), education, regional child care availability and regional dummies. Moreover to identify the selection correction term in the wage equation we exclude from the wage equation child care availability and all other variables relating to children. Since we use six cross sections over time our instruments display both time series and regional variation.

III. DATA AND EMPIRICAL RESULTS

III.1. *The UK Tax-Benefit System and the Distribution of Hours*

The UK tax and benefit system consists of three main components: income tax, National Insurance contributions and the benefit system, which itself includes over twenty separate benefits many of which are means-tested. For many single parents, particularly on low earnings levels, the effect of the tax system on their budget constraint is complex. However, for a large range of earnings the tax system is flat with a constant marginal rate applicable to most taxpayers.

The FES provides information on taxes paid and benefits received by each individual in the sample, from which we construct marginal tax rates. Thus we avoid the problem of modelling benefit take-up when marginal rates are calculated on the basis of entitlement. A description of the data used in our study is provided in the Data Appendix; one third of the working sample work on the 'flat' part of the tax system. As we can see from Fig. 1, the distribution of hours for the 'tax selected' sample is very dispersed in contrast to other European countries (see Bourguignon and Magnac (1990) for example).

To motivate the specification of our model we present in Fig. 2 two Kernel regressions of hours on the wage rate.[8] The top regression line relates to a sample of individuals with other income below the median while the bottom regression to individuals with other income above the median.[9] The results

[8] These regressions were performed using an interactive Kernel regression program by Duncan and Jones (1991).

[9] We have also performed multivariate Kernel regressions over the entire sample rather than this crude split in two groups. Results were similar, but the graphs presented in Fig. 2 summarise the findings in a clearer way.

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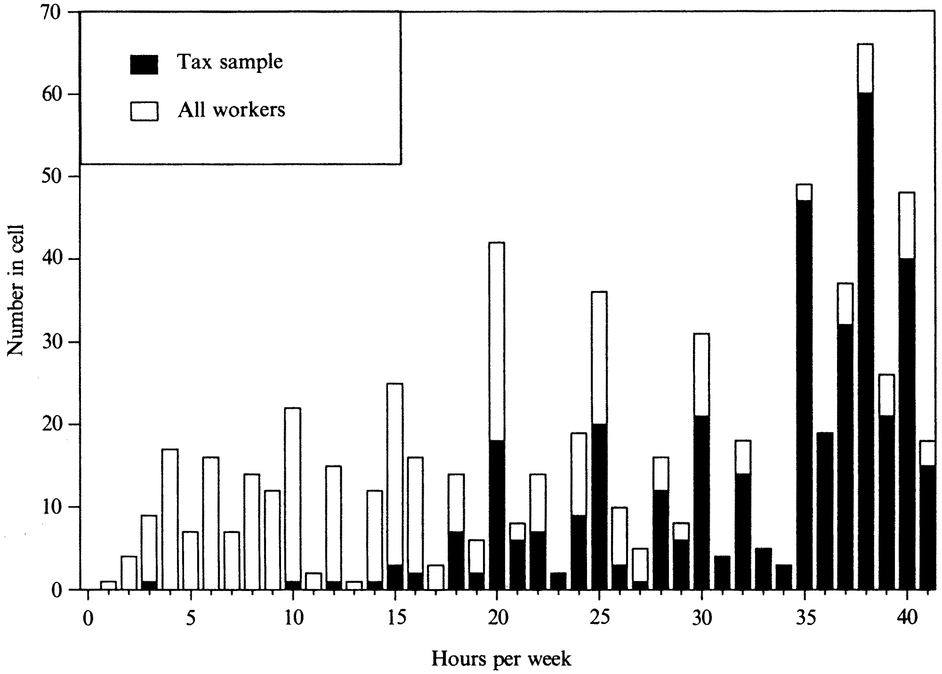

Fig. 1. Hours distribution.

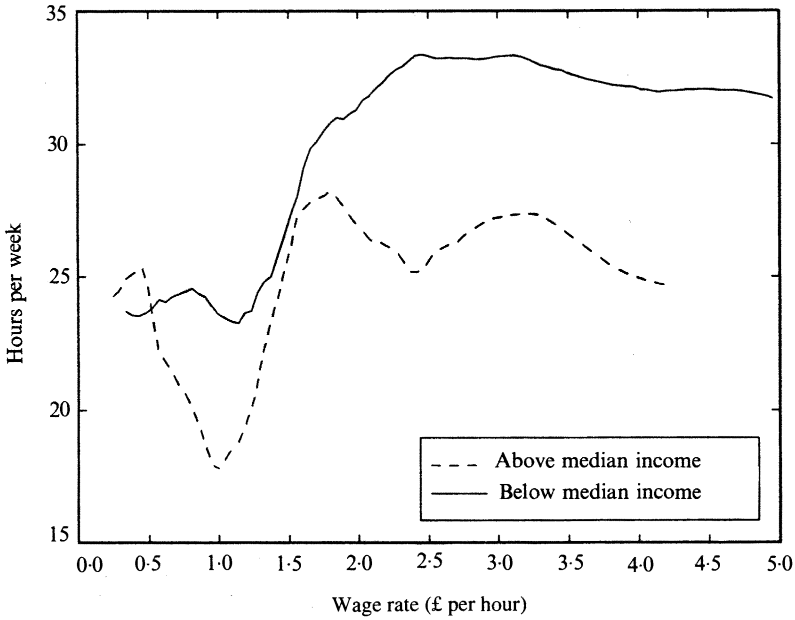

Fig. 2. Kernel regression: 1981–6 data.

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should be interpreted with caution since no allowance is made for endogeneity of the wage or other income. Moreover note that below a wage of £1.00 the data are sparse. Given the above, the data reveal a clear non-linear relationship between hours and wages with labour supply being forward sloping at lower hours (initially quite steeply so) and then becoming backward bending. Moreover the positions of the two graphs reveal a strong negative income effect on hours. On this basis it is important to use a model that allows for such nonlinearities. The Heckman (1974) model is useful in this respect.

III.2. *Labour Supply Estimates and Elasticities*

In the Appendix we provide the reduced form participation equation for the work/non-work split.[10] We also provide a reduced form log marginal wage equation and a reduced form other income equation. These allow us to correct for endogeneity in the marginal wage and other income and for sample selection. Note the regional child care variables, drawn from *Regional Trends*, in the participation probit (Appendix, Table A1): these are the regional child care density interacted with a dummy for the presence of children aged 0–2. The same variable is then interacted with the dummy for children in the 3–4 age group that are not yet eligible to go to school. We also interact the child care availability variable with children aged 3–4 at school and with children in primary school and in secondary school. When we exclude time dummies (which are not significant) from the participation equation, the coefficients of the first two variables (relating to pre-school children) do not change but their standard errors are halved implying that the presence of child care facilities significantly raises the participation of lone mothers with pre-school children. These results have a clear interpretation in terms of the effects of fixed costs of work and could have interesting policy implications.

The first column of Table 1 provides estimates of the 'tax selection' model TS. Household composition is modelled by including zero-one dummies DKi, $i = 1, \ldots, 4$, to reflect the presence of the youngest child in each age range 0–2, 3–4, 5–10 and 11+. The parameter ρ is the correlation between the selection equation and the unobservable component of the structural equation entering through α. The demographic composition variables are not significant, which is not surprising since women with young children are highly under-represented in our selected sample since they mainly work low hours. Moreover older children have relatively little effect on labour supply. Note though that demographics still affect labour supply through μ which reflects intertemporal consumption choice. Both the β and the γ parameters are positive as predicted by the theoretical discussion, and hence guarantee global integrability.

On the statistical properties of the model note that the other income residual ϵ_μ is significant implying endogeneity of other income. Moreover, once we have corrected for tax rate endogeneity by our selection procedure the wage rate can be treated as exogenous; the exogeneity test for the wage is 0·46 (asymptotically

[10] Due to space limitations, we do not present the reduced form equation which controls for selection onto the flat part of the budget constraint.

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Table 1
Labour Supply Estimates (1981–6 Data)

Variable	(1) TS	(2) CS	(3) GM
α	−9·751 (3·011)	−6·430 (1·300)	−11·02 (4·654)
α (*DK*2)	0·563 (0·697)	1·222 (0·407)	0·181 (0·933)
α (*DK*3)	0·148 (0·498)	1·397 (0·401)	−0·168 (0·766)
α (*DK*4)	−0·367 (0·558)	0·905 (0·385)	−0·562 (0·817)
α (*AGE*)	0·313 (0·254)	0·185 (0·129)	0·322 (0·371)
β	0·254 (0·090)	0·140 (0·032)	0·313 (0·143)
γ	0·015 (0·005)	0·010 (0·004)	0·012 (0·006)
σ	2·039 (0·608)	2·021 (0·434)	2·383 (0·997)
ρ_λ	0·072 (0·181)	0·711 (0·093)	0·258 (0·222)
ρ_w	—	0·226 (0·118)	—
ρ_μ	−0·011 (0·005)	−0·006 (0·004)	−0·007 (0·007)
Log likelihood	−2,037·08	−3,751·12	−1,529·44
Skewness (D.F. = 1)	4·428	23·16	—
Kurtosis (D.F. = 1)	1·543	0·349	—
Normality (D.F. = 2)	4·551	68·81	—

Notes: TS: Model estimated on the flat part of the budget constraint. CS: Model estimated on the complete sample. GM: Model estimated by grouping around 20 and 40 hours and on the flat part of the budget constraint. Standard errors are listed in parentheses. The normality test statistics reported here follow the Generalised Residual methodology of Gourieroux *et al.* (1987) and Chesher and Irish (1987). Reduced form estimates are presented in Appendix Table A1. Default child group is *DK*1.

N(0,1)). Finally note that the diagnostics presented at the foot of the table do not indicate misspecification for the TS model.

We now proceed with two additional empirical experiments. First we reestimate the same model using the entire sample of working lone parents. This is a 'conventional' labour supply model with marginal after-tax wage rates. The results are presented in column 2 (model CS) and indicate that the CS model is misspecified since the parameter estimates obtained by this more restrictive procedure are quite different. In particular the wage coefficient β is much lower. Moreover the wage exogeneity test now rejects strongly revealing that the tax system is the main source of endogeneity for after-tax wages. Finally the normality test statistics also indicate misspecification. An interpretation of these differences is that the instruments used in the reduced form wage equation are not sufficient to control for the endogeneity and/or measurement error in tax rates. The cost of our procedure is reflected in the evident loss of precision. It should be stressed that the CS results are restrictive not only due to the treatment of taxes but also because we are fitting the same labour supply equation over a much wider range of data. In fact our selectivity procedure could just have had the effect of relaxing the preference structure.

The second empirical experiment we carry out relates to hours constraints. The assumption of hours flexibility underlies most empirical studies of labour supply. The type of data available does not allow for a convincing model of the alternative but we can carry out an informal test. In column 3 (GM model) we present results from estimating our model on the tax selected sample but where those individuals working at 19–21 hours and at 38–40 hours do not contribute

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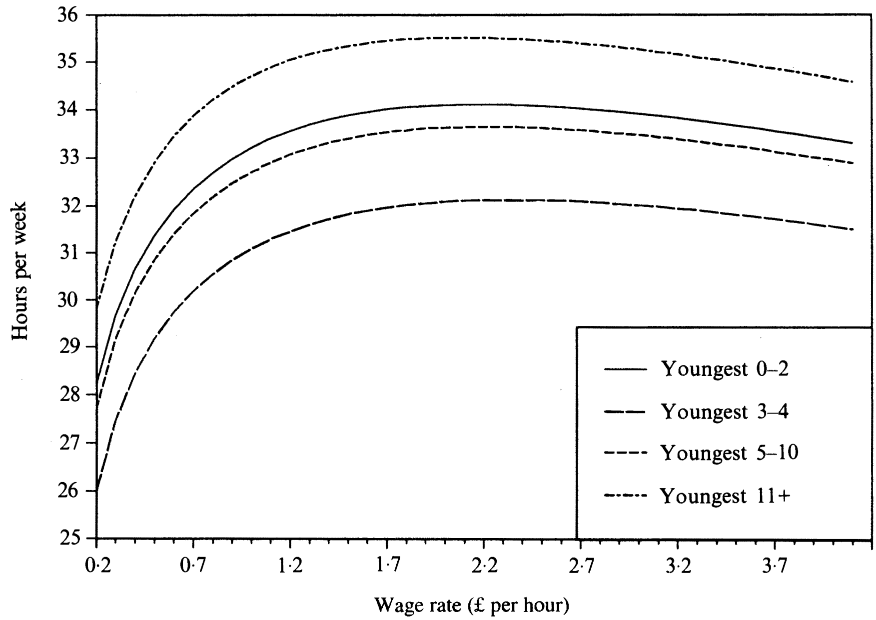

Fig. 3. Estimated implicit hours function TS.

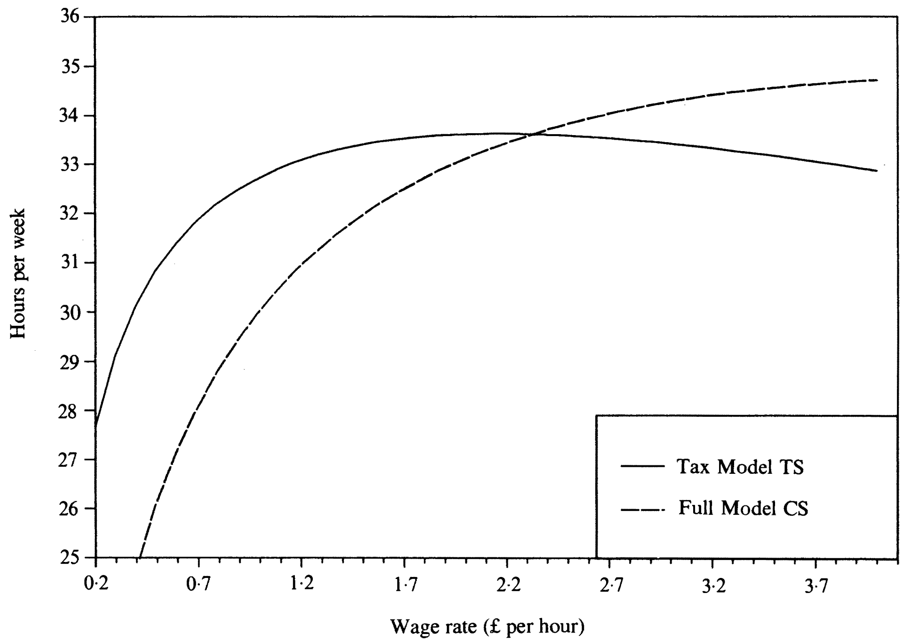

Fig. 4. Comparison of models TS and CS.

[11] This is just a simple specification test and its power against the alternative we are interested in depends on there being some groups in the population who can freely choose their hours of work. Here we are implicitly assuming that individuals are constrained in the neighbourhood of the standard full time and the standard part time hours and that most of the remaining individuals are on their labour supply curve. If this assumption is invalid our test may not have much power.

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Table 2
Distribution of Elasticities

	Pre-school			Primary school		
	TS	CS	GM	TS	CS	GM
e (min)	−0·09	−0·04	−0·05	−0·14	−0·10	−0·08
e (25)	0·01	0·07	0·02	0·04	0·12	0·05
e (50)	0·08	0·17	0·07	0·10	0·22	0·09
e (75)	0·25	0·45	0·21	0·23	0·44	0·19
e (mean)	0·26	0·52	0·23	0·18	0·38	0·16
S.D.	0·55	0·99	0·45	0·25	0·45	0·20
	Secondary school			**All workers**		
	TS	CS	GM	TS	CS	GM
e (min)	−0·17	−0·09	−0·10	−0·17	−0·10	−0·10
e (25)	−0·01	0·06	0·01	0·01	0·07	0·02
e (50)	0·06	0·14	0·06	0·07	0·18	0·07
e (75)	0·13	0·26	0·11	0·18	0·36	0·15
e (mean)	0·11	0·26	0·11	0·16	0·34	0·14
S.D.	0·25	0·44	0·20	0·31	0·55	0·25

e (25), e (50) etc., elasticity at the 25th, 50th percentile, etc.

to the hours of work density. For them we just assume that desired hours of work are positive. The likelihood contributions for these observations become $\int_{h>0} f(h)\,\Pr(I>0\,|\,h)\,dh$ instead of $f(h)\,\Pr(I>0\,|\,h)$.[11] As can be seen by comparing these results with the TS model, neither the wage nor the other income coefficient changes much; as we expect the grouping procedure involves just some loss in precision. In Fig. 3, we plot the estimated implicit hours function TS split by the age of the youngest child. All other variables are set at their group means.

Fig. 4 compares the results of the TS with the CS model.The former captures a greater degree of backward bending labour supply behaviour. In Table 2 we also compare the distribution of elasticities. The average elasticity for the TS model is half that of the CS model. These numbers show that lone parent labour supply can be quite wage elastic at low hours.

IV. CONCLUSIONS

The purpose of this paper has been to develop an empirical methodology for estimating labour supply models in the presence of taxes. Our approach respects the coherency conditions originating from the structure of the budget constraint without imposing restrictions on the economic model. We use our method to analyse the labour supply of lone parents, an issue with important policy implications. Our empirical findings can be summarised as follows.

1. Accounting for non-linear taxes in the way we suggest changes the results significantly. Most of the endogeneity of after-tax wages comes from the

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endogeneity (or measurement error) in the marginal tax rates at low hours where the benefit system is active.

2. The labour supply behaviour of lone parents is quite wage elastic at low hours and becomes flat and slightly backward bending at higher wage rates.

3. The integrability conditions for labour supply are supported by the data.

4. There is little evidence of hours constraints. When bunching around 20 and 40 hours of work is accounted for, labour supply estimates do not change significantly.

5. Finally it has been illustrated that it is very important to use a specification for preferences that can capture the evident non-linearities in the data.

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Data appendix

Sample selection: (a) Age at which education ceased < 25, (b) No self-employed, HM Forces or retired, (c) Northern Ireland excluded.

Descriptive Statistics

Family Expenditure Survey, Lone Mothers

Year of sample	1981	1982	1983	1984	1985	1986
Participation rate	0·479	0·451	0·371	0·409	0·396	0·394
Marginal wage (workers)	1·709	1·373	1·691	1·644	1·593	1·610
	(1·165)	(0·958)	(1·233)	(1·278)	(0·992)	(1·087)
Other income	67·955	65·894	64·559	73·104	69·986	72·062
	(56·045)	(59·397)	(63·465)	(53·348)	(55·685)	(56·768)
Youngest child < 2	0·185	0·214	0·237	0·239	0·226	0·286
Youngest child 2–4	0·104	0·117	0·113	0·108	0·132	0·161
Youngest child 5–10	0·390	0·338	0·343	0·317	0·347	0·270
Youngest child 11 +	0·320	0·331	0·307	0·336	0·294	0·283
Age	35·081	34·120	33·102	33·822	33·902	32·845
	(8·853)	(9·693)	(8·661)	(8·793)	(8·641)	(8·971)
Age left education	15·737	15·444	15·728	15·807	15·921	16·102
	(1·631)	(1·145)	(1·417)	(1·728)	(1·561)	(1·672)
Sample size	259	266	283	259	265	322
Tax selected sample	124	120	105	106	105	127

Note: Standard errors are listed in parentheses.

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Table A1
Reduced Form Equations

	Participation		Log marginal wage		Other income	
Variable	Coeff.	S.E.	Coeff.	S.E.	Coeff.	S.E.
Constant	−0.372	1.379	4.778	0.337	−5.995	61.57
Age	−0.078	0.079	0.246	0.059	13.329	3.45
Age^2	−0.146	0.055	0.012	0.043	4.497	2.30
Education	0.216	0.052	0.176	0.043	−1.652	2.36
$(Education)^2$	−0.010	0.007	−0.005	0.005	−0.259	0.30
(Age) × (Education)	0.008	0.030	−0.018	0.020	−1.346	1.20
Number of children	−0.222	0.047	—	—	10.231	1.93
*DK*1	−1.220	0.305	—	—	30.398	13.38
*DK*2(*NS*)	−1.524	0.525	—	—	−9.630	19.68
*DK*2(*S*)	−1.039	0.594	—	—	7.232	20.43
*DK*3	−0.036	0.245	—	—	2.761	11.70
North	0.720	0.990	−0.973	0.317	61.013	43.89
Yorkshire	0.569	0.856	−0.802	0.307	50.512	37.72
North West	0.170	0.503	−0.721	0.277	34.793	22.11
East Midlands	0.334	0.670	−0.648	0.240	36.367	29.32
West Midlands	0.004	0.509	−0.501	0.282	26.274	22.16
East Anglia	0.458	0.518	−0.458	0.157	22.428	23.24
South West	0.287	0.584	−0.327	0.140	37.342	25.45
Wales	0.835	1.042	−0.337	0.241	73.413	47.33
Scotland	0.851	0.954	−0.533	0.203	63.194	42.84
Female unemp/age	−1.128	1.936	−0.311	1.616	−15.243	81.98
Female unemp/region	−3.732	4.662	3.367	3.569	−51.092	209.71
Vacancies/region	−0.059	0.172	−0.192	0.132	4.086	7.38
Redundancies/region	0.020	0.041	0.045	0.032	−3.421	1.81
Childcare × *DK*1	2.343	2.145	—	—	94.503	95.83
Childcare × *DK*2(*NS*)	3.525	2.222	—	—	156.363	99.56
Childcare × *DK*2(*S*)	2.485	2.273	—	—	135.841	100.55
Childcare × *DK*3	1.744	2.109	—	—	133.365	95.25
Childcare × *DK*4	2.156	2.124	—	—	130.545	95.95
Year = 1982	−0.001	0.170	−0.070	0.132	−12.838	7.46
Year = 1983	−0.213	0.288	0.286	0.219	−21.524	12.71
Year = 1984	−0.191	0.416	0.249	0.295	−25.614	17.88
Year = 1985	−0.277	0.470	0.272	0.294	−34.573	20.04
Year = 1986	−0.687	0.726	0.697	0.420	−43.507	32.05
Selectivity correction	—	—	−0.021	0.117	—	—

Notes: Child care availability is measured by the number of full child care places per 1,000 of the regional population (source: *Regional Trends* 1981–1986). *DK*2(*NS*) indicates the presence of the youngest child aged 2–4 not at school. *DK*4 is the default child group. Vacancies and redundancies by region are taken from *Employment Gazette* 1981–1986. Participation: dummy = 1 for individual working, 0 elsewhere. This relates to the CS model. Selection equations for TS, GM models are not presented here.

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